

ORACLE

Cloud Operations for Exadata Database Service on Exascale Infrastructure

Management Interfaces

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Office of CTO | EMEA Technology Engineering
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Agenda

- Introduction
- Service Overview
- Management Interfaces
 - Oracle ExaCC Control Plane based: UI, OCI-CLI, REST API, SDK, DevOps Tools
 - Guest VM Command Line Interface (CLI): dbaascli

Introduction

Introduction



This Workshop provides information about Exadata Database Service on Exascale Infrastructure and its different components, typical operations, and maintenance activities to be performed in the platform by different automation practices

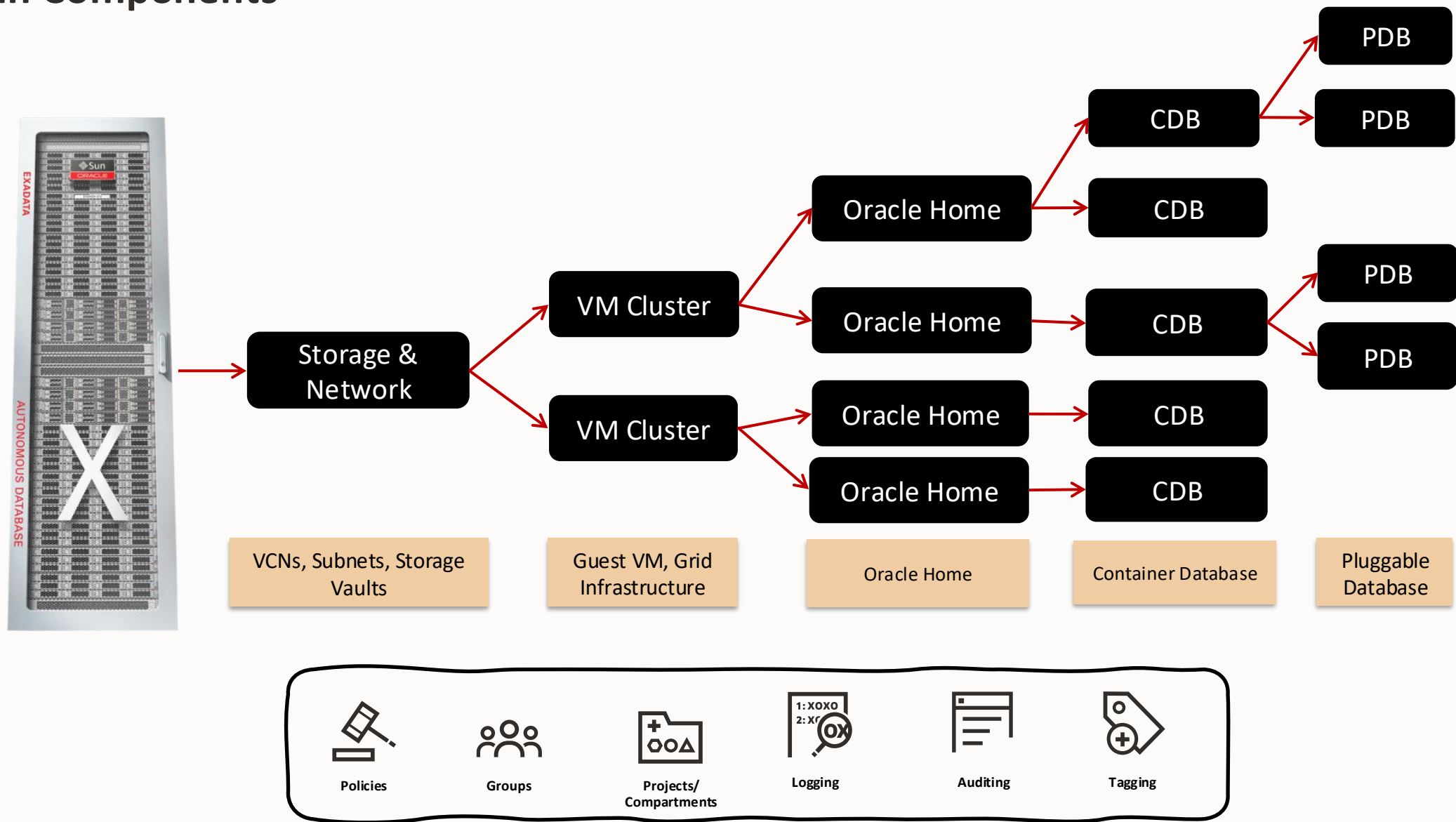
We'll cover different aspects of the infrastructure in terms of service architecture, management interfaces, the information needed to deploy a VC, OH, DBs, DG, scaling and patching

The code in this Workshop is just sample code and can be incomplete.

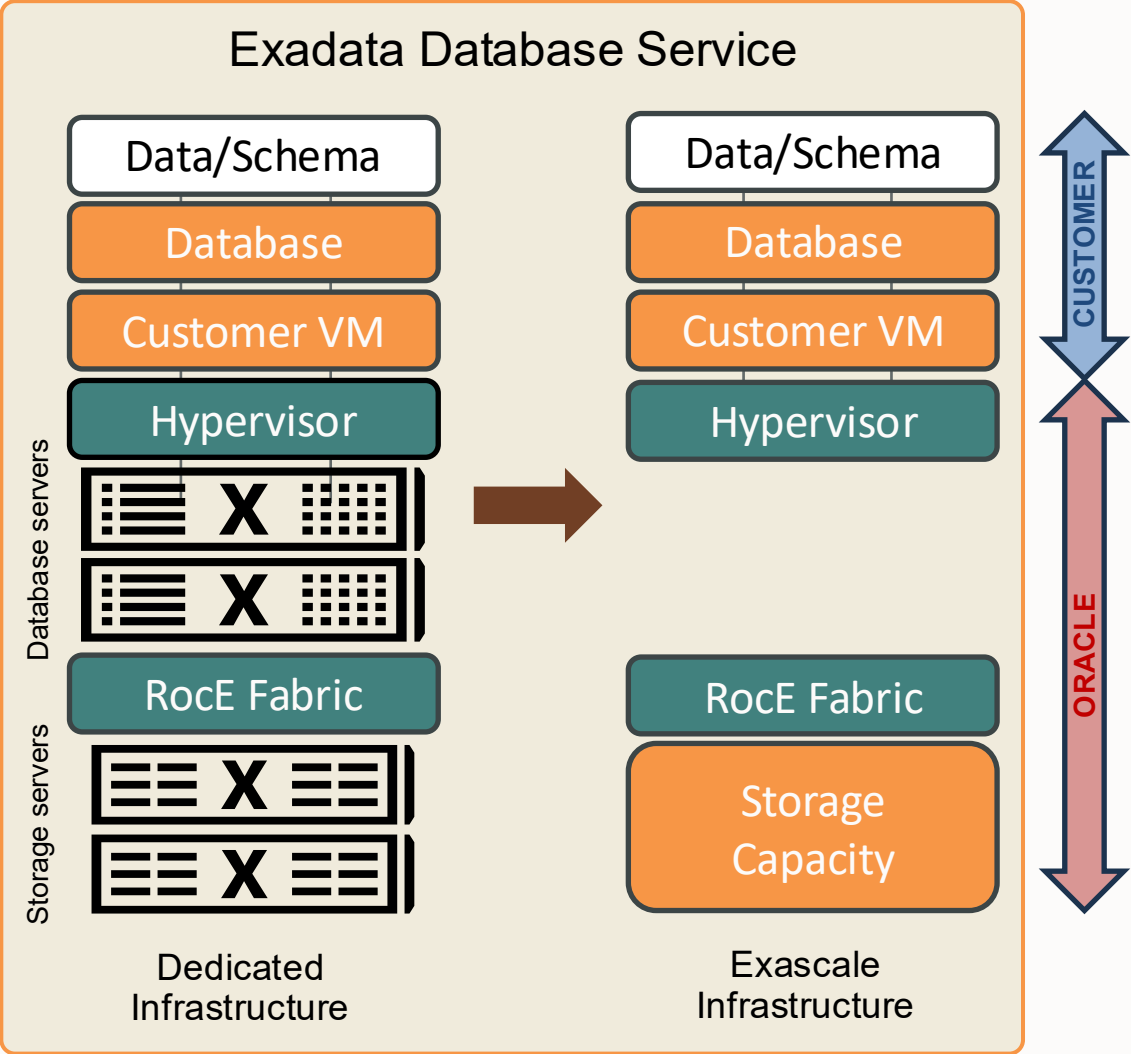
To make the sample code work in your Oracle Cloud tenancy, please replace the values for any parameters whose current values do not fit your use case and check the official documentation to include missing parameters.

Service Overview

Main Components



Roles and Responsibilities



Function	Oracle Cloud Ops	Customer
Monitoring	Infrastructure, Control Plane, Availability, Capacity	Customer OS, Databases, Schemas
Incident Management & Resolution	Support for any incident related to the underlying platform	Incident Management and resolution for customer's Apps
Patch Management	Proactive patching of infrastructure	Patching of tenant instances and testing
Backup & Restoration	Provide running and customer accessible VMs.	Backup & Recovery of customer's data using Oracle native or 3 rd party capability
Cloud Support	Response & Resolution of SRs	Submit SRs via MOS/Support Portal



Management Interfaces

Management Interfaces



The Exadata Cloud resources on Oracle Cloud Infrastructure (OCI) are created and managed through a variety of interfaces provided to fit your different management use cases:

Oracle Cloud Web based: UI, OCI-CLI, REST API, SDK, DevOps Tools

- The Exadata Cloud UI uses Rest APIs under the covers

Guest VM Command Line Interface (CLI) eg dbaascli

- Local to the guest VMs (no reliant on Control Plane connection)
- Tool actions are refreshed in the UI every 10 minutes.

Best practices on how these interfaces align for various Exadata Cloud database management use cases: Cloud API/CLI Alignment Matrix (Doc ID 2768569.1)

ExaDB-XS Automation Highlights



Virtual Machine Cluster

- Create, delete, start, stop, restart
- Add/remove VM Cluster nodes
- Memory, local storage, Vault Storage scaling
- ECPU scaling
- Patch apply, patch rollback
- Upgrade
- (...)

Database and Grid Infrastructure

- Create, delete CBDs/PDBs
- Start, stop, restart CBDs/PDBs
- Backup, restore
- Patch apply, patch rollback
- Upgrade
- Move DB to a different Oracle Home
- Clone PDBs
- Data Guard setup, switchover, failover
- Transparent Data Encryption (TDE)
- (...)

Management Interfaces

Control Plane Interfaces

OCI Control Plane Interfaces

OCI APIs

REST APIs that use HTTPS requests and responses for managing OCI resources, including Exadata Cloud

OCI Console

An intuitive, graphical interface for creating and managing your Exadata Cloud and other OCI resources, is one of the interfaces to the OCI APIs

Command Line Interface (OCI CLI)

Lightweight tool for command-line tasks. Supports scripts and extends Console functionality

Software Development Kit (SDK)

Enable custom solutions for Exadata Cloud and OCI services

DevOps Tools & Plug-ins

Tools like Terraform Provider and Ansible Collection for provisioning and automation

Cloud Shell

Browser-based terminal with pre-authenticated CLI, useful for tutorials and quick commands

OCI Console

Navigation Menu → Oracle Database → Oracle Exadata Database Service on Exascale Infrastructure

☰ Cloud

Oracle Database

Oracle Exadata Database Service on Exascale Infrastructure

VM Clusters

Exascale Storage Vaults

Resources

Software images

Standalone backups

Search resources, services, documentation, and Marketplace

Germany Central (Frankfurt)

VM Clusters

The VM Cluster is where Oracle Databases are deployed.

Search and Filter

Search

Applied filters

Compartment Shared Specialists

Create VM Cluster

Display name	State	VM count	ECPU enabled	Memory (GB)	Exascale Storage Vault	Created	
MAM19c	Available	2	16	66	mam19c	Tue, Sep 23, 2025, 14:43:10 UTC	...
snexaxs	Available	2	32	220	Vault-snban	Mon, Aug 25, 2025, 04:59:36 UTC	...
RW-EDXS	Available	2	16	110	RWVAULT	Tue, Sep 10, 2024, 11:52:29 UTC	...
MB-Clust	Available	2	0	44	MB-Vault-202407311218	Wed, Jul 31, 2024, 11:23:19 UTC	...

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Command Line Interface (OCI CLI)

Installation

- Requirements:
 - ✓ Oracle Cloud infrastructure account
 - ✓ User created in that account with needed permissions
 - ✓ Keypair for signing API requests with public key uploaded to Oracle (can use token-based authentication instead)
 - ✓ Supported version of Python on the operating system where the CLI would be installed
 - ✓ User must have enough IAM policies granted for the resources it will manage
- Follow the steps for your operating system from [here](#)

Configuration

```
$ oci setup config
```

Before using the CLI, create a config file with required credentials

- User OCID
- Tenancy OCID
- Region
- RSA signing key
 - Upload your RSA public key into the console → Profile menu → User Settings → Add Public Key

NOTICE: The API Signing Key can not be used with the federated users

Command Line Interface (OCI CLI)

Syntax

```
$ oci [OPTIONS] COMMAND [ARGS] ...
```

Where COMMAND usually translates for:

```
<service> <type> <action> <options>
```

Help

```
$ oci --help
$ oci db -h
$ oci db system -h
```

Use environment variables

```
$ export T=<tenancy OCID>
$ export C=<compartment OCID>
$ oci iam compartment list -c $T
```

Authentication

```
$ oci session authenticate
$ oci iam region list --profile <profile> --auth
security_token
```

A web browser window will be open asking for OC console login. Once we've entered our user details the windows could be closed

Input

```
$ oci [OPTIONS] COMMAND [ARGS] --from-json [text] ...
```

Inputs with more that one value, are passed in JSON format and can be provided as a string at the command line or as a file.

The **--generate-full-command-json-input** option can be used to generate a sample json file to be used with this command option

Output format

```
$ oci iam region list
or
$ oci iam region list --output table
```

By default, all responses are returned in JSON format. Depending on the output, it could be better to ask for a table format output with "--output table"

Software Development Kits (SDK)



The Oracle Cloud Infrastructure (OCI) [Software Development Kits](#) provides a programmatic way to manage Exadata Database Service on Exascale Infrastructure resources. It supports multiple programming languages, making it flexible for automation and integration needs.

- Multi-Language Support:
 - [Java](#), [Python](#), [Typescript/Javascript](#), [.NET](#), [Go](#), [Ruby](#), [PL/SQL](#)
- Common Use Cases:
 - Develop code to provision or change resources on OCI.
 - OCI SDK is available on your corporate language.
 - You're using a IaC procedural operating model.
- Resources:
 - SDK Documentation: [Oracle Cloud SDKs](#).
 - [Example Code](#): Available on GitHub for each language.

```
>>> import oci
# Set up config
>>> config = oci.config.from_file(
...     "~/.oci/config",
...     "DEFAULT")
# Create a service client
>>> identity = oci.identity.IdentityClient(config)
# Get the current user
>>> user = identity.get_user(config["user"]).data
>>> print(user)
{
  "compartment_id": "ocid1.tenancy.oc1...",
  "description": "Test user",
  "id": "ocid1.user.oc1...",
  "inactive_status": null,
  "lifecycle_state": "ACTIVE",
  "name": "test-user@corp.com",
  "time_created": "2016-08-30T23:46:44.680000+00:00"
}
```

DevOps Tools and Plug-ins

Infrastructure as Code (IaC) - Possibilities



Terraform can be used to provision VM Cluster Networks, VM Clusters, DB-homes, databases, etc., and perform some changes (scaling)

The Exadata infrastructure resource must be taken carefully, as in an element configured by Oracle Installation Services. The resource can be discovered and create a terraform configuration, but any change/mistake that forces the recreation of the infrastructure could be a serious problem

From a IaC perspective, Terraform is not enough to cover the whole lifecycle, as there are some operations that can not be managed with it, as patching the OS/Grid/DBs

Terraform has some special resources to handle other types of operations (null resource, remote- exec), but other solutions do it better, as **Ansible**. We say that these operations follow a Procedural Pattern, in contrast with the Declarative Pattern used by Terraform

The use of Ansible let us to have playbooks with the steps of the operation configurable by the developer/operator and having some code reusable

DevOps Tools and Plug-ins

Infrastructure as Code (IaC) - Terraform

Terraform is the most popular tool used in Infrastructure as Code (IaC) environments to deploy and change infrastructure based on HashiCorp Configuration Language (HCL).

It is a declarative language where you define the resources that you want to provision, and it takes care of the precedence and requirements between resources.

OCI has a provider: [Oracle Cloud Infrastructure Provider](#)

Terraform can be used to provision the following resources in Exascale:

- Exascale on Cloud exadata infrastructure: [oci_database_cloud_exadata_infrastructure_configure_exascale_management](#)
- Exadata VM cluster on Exascale Infrastructure: [oci_database_exadb_vm_cluster](#)
- Exascale DB Storage Vault: [oci_database_exascale_db_storage_vault](#)
- DB Homes: [oci_database_db_home](#)
- Database: [oci_database_database](#)
- Others...

Terraform is not enough to carry on all activities. Maintenance operations (as patching), can not be done with Terraform and usually you need other configuration management tools as Ansible, scripting...

DevOps Tools and Plug-ins

Infrastructure as Code (IaC) - Resource Manager



What is Terraform?

Terraform is an open-source infrastructure as code software tool used for building, changing, and versioning infrastructure safely and efficiently.

It enables users to define and provision a datacenter infrastructure using a high-level configuration language known as Hashicorp Configuration Language (HCL).

- No TF State File Management Control
- Multiple users can modify same resources simultaneously
- Each user must maintain their credentials; sometimes hard-coded

What is OCI Resource Manager?

Resource Manager is an OCI managed service that allows you to automate the process of provisioning your Oracle Cloud Infrastructure resources.

It uses Terraform to codify your infrastructure in declarative configuration files, which allows you to review and edit, version, persist, reuse, and share them across teams.

- Terraform state file is stored / managed by OCI
- Only one action is allowed at a time for resource collections defined as Stacks
- Permissions controlled by IAM

DevOps Tools and Plug-ins

Infrastructure as Code (IaC) - Resource Manager

Stacks

- Stacks represent definitions of groups of Oracle Cloud Infrastructure resources that you can act upon as a group

Configurations

- A configuration is a set of one or more Terraform configuration (.tf) files that specify the Oracle Cloud Infrastructure resources in a given stack
- Each stack has a configuration, which is made up of one or more declarative configuration files

Jobs

- Jobs perform the actions that are defined in your configuration
- You can Plan, Apply or Destroy a job



**Terraform on OCI with
Resource Manager:**
<https://youtu.be/w3no1zaZgxA>

DevOps Tools and Plug-ins

Infrastructure as Code (IaC) - Resource Manager



- OCI Ansible Collection: <https://github.com/oracle/oci-ansible-collection>
- Supported Services: For the complete list of supported services, please check the modules list.
- Ansible Tower: OCI Ansible Collection supports Ansible Tower and AWX.
- Ansible Inventory Plugin: enables the tracking configuration resources by preserving lists, called inventory lists, as simple files. These inventory lists can be static or dynamic. Dynamic lists can automatically update when inventory resources are added, deleted, or moved.
- The OCI Ansible collection requires Ansible version 2.9 or later.

DevOps Tools and Plug-ins

Infrastructure as Code (IaC) - Resource Manager

Automation

Resource Manager automates the process of provisioning your resources with Terraform

Built-in security

Integrates with IAM so that developers can define granular permissions for infrastructure operations

Stacks

Maintains the collection of OCI resources corresponding to a given Terraform configuration. Jobs are created when you run terraform actions

Templates

Create terraform configurations for private use or share them with others in your tenancy

State file

Resource Manager generates and updates state files automatically when you run jobs. It also supports state locking by allowing only one job at a time to run on a given stack

Integration with CI/CD

Easily integrated with CI/CD systems (configuration source providers), such as GitLab

Discovery

Enables you to duplicate or rebuild existing infrastructure

Drift detection

The **drift detection** feature allows to detect deviations in the current infrastructure configuration from the desired configuration

Cloud Shell

What is Cloud Shell?

- A browser-based terminal in Oracle Cloud Console.
- Provides a pre-configured Linux shell.
- Includes Oracle Cloud Infrastructure (OCI) CLI, Ansible, and other tools.
- Free to use (within monthly tenancy limits).

Key Features

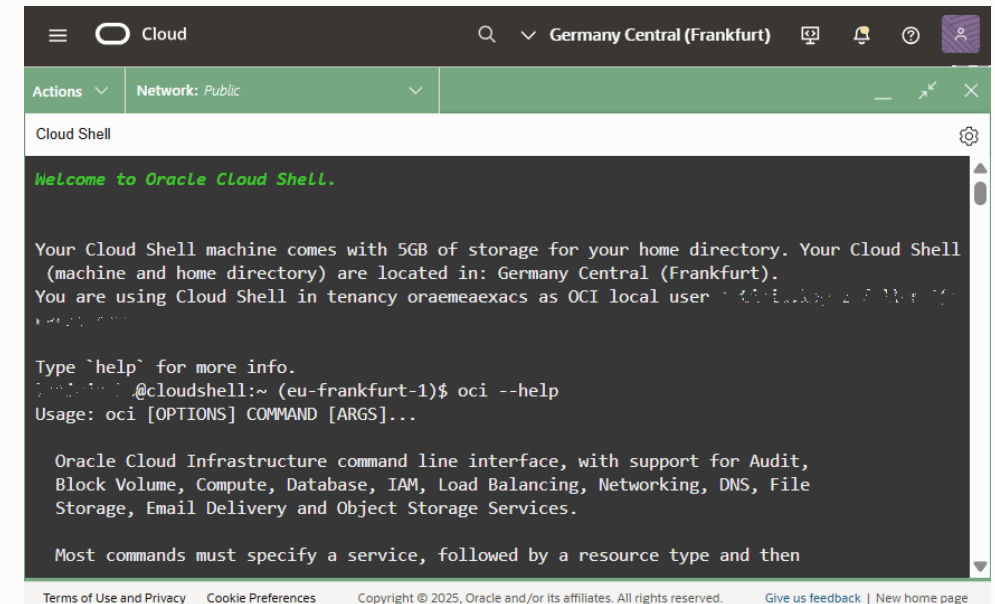
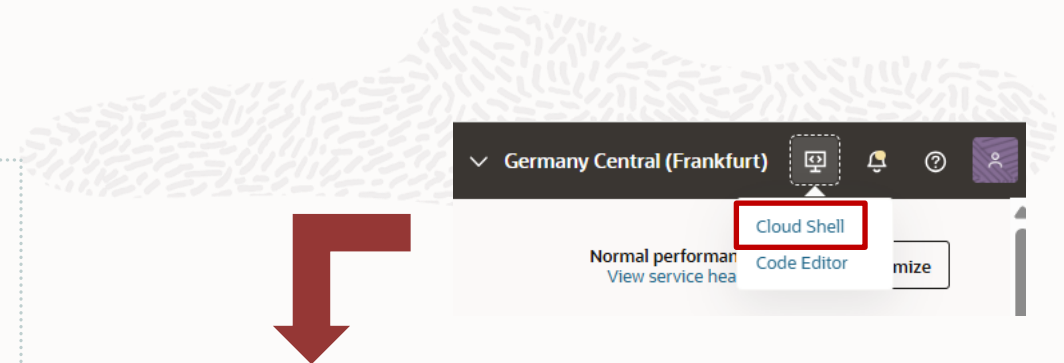
- Ephemeral VM for running commands.
- 5GB of encrypted persistent storage.
- Persistent frame stays active while navigating the Console.
- Accessible directly from the Console, no setup required.

Access Control

- IAM policy required for access.
- Can access OCI resources based on granted permissions.
- Public internet access only with proper security policies in place.
- Network access limited by default to internal tenancy resources.

Limitations

- No root access or sudo commands.
- Maximum session length: 24 hours (auto logoff after 60 minutes of inactivity).
- 5GB storage persistent but may be removed after 6 months of inactivity.
- No external connections (e.g., SSH access not allowed).
- Cannot mount additional storage or use ping command.
- Works in UTC time zone, no ability to change.



Pre-installed Tools:

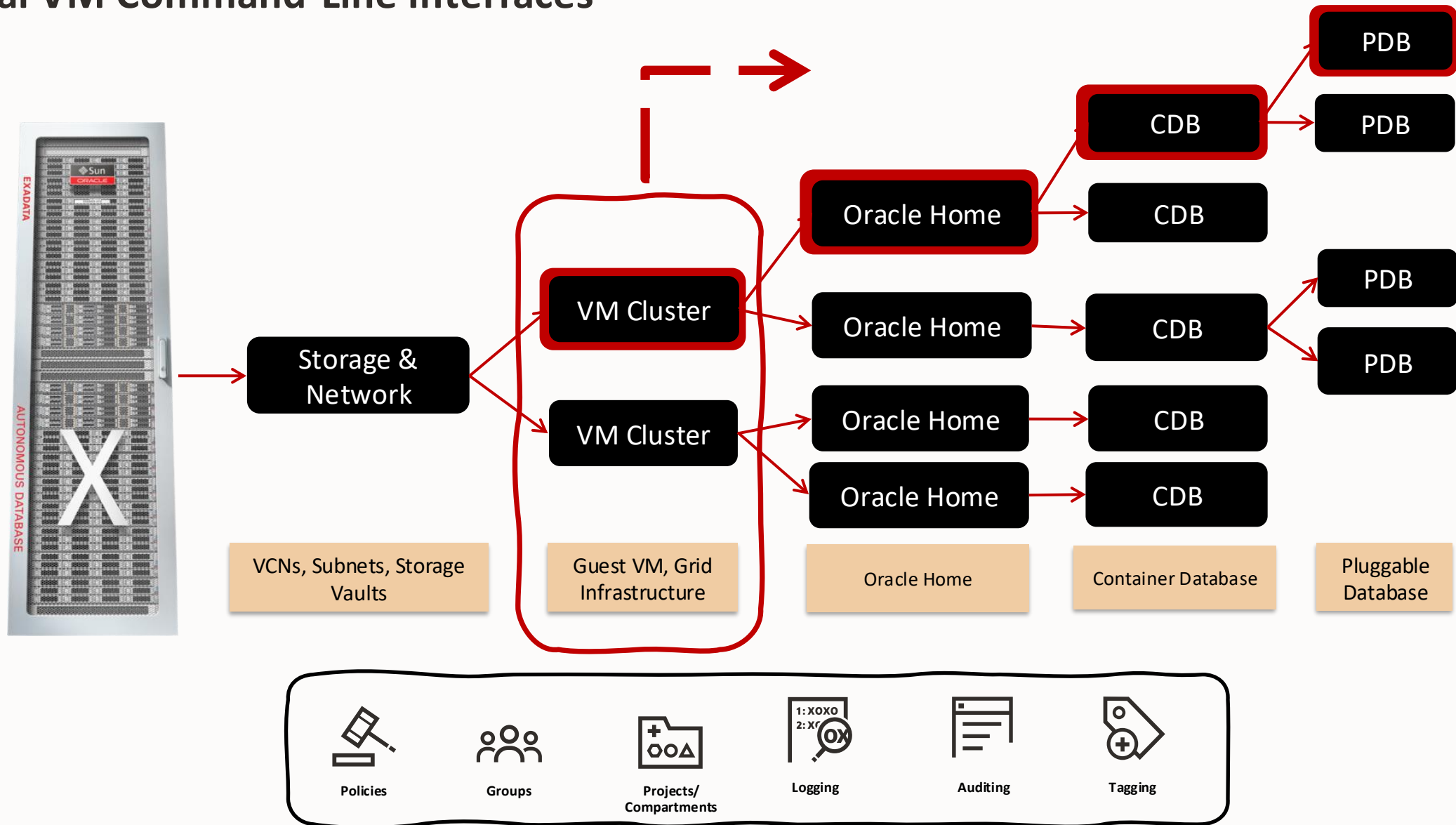
- OCI CLI, Git, Java, Python, Terraform, Ansible, Kubernetes tools (kubectl, helm), Docker, and more.
- Useful SDKs: Java, Python, Go, TypeScript, etc.



Management Interfaces

Local VM Command-Line Interfaces

Local VM Command-Line Interfaces



Local VM Command-Line Interfaces

CLI utilities located on the VM guests

Provisioned as part of the VM clusters on the Exadata Cloud Infrastructure, are available to perform various lifecycle and administration operations

dbaascli

Use the dbaascli utility to perform various database lifecycle and administration operations on the Oracle Exadata Database Service on Exascale Infrastructure such as changing the password of a database user, starting a database, managing pluggable databases (PDBs), scaling the CPU core count in disconnected mode, and more.

bkup_api

Handles backup/recovery tasks, including on-demand database and PDB backups.

DEPRECATED

Must be connected to a VM in the Exadata Cloud VM cluster; uses VM OS user security, not OCI user security.

Changes sync to OCI Control Plane via **DB Sync**.
Tools are auto-updated by Oracle on a regular basis.

dbaascli operations



- Administration and Configuration
- Cloud Software Library (CSWLIB) Management
- Database Management
- Data Guard Management
- Database Home Management
- Diagnostics and Health Checks
- Grid Infrastructure Management
- Patching and Upgrading
- Pluggable Database (PDB) Management
- System Management
- Transparent Data Encryption (TDE) Management

Reference: <https://docs.oracle.com/en-us/iaas/exadb-xs/doc/exadb-xs-using-dbaascli.html>

dbaascli examples

List all commands

```
# dbaascli list [-help]
```

Show latest dbaastool available

```
# dbaascli admin showLatestStackVersion
```

Install/update a dbaastools RPM

```
# dbaascli admin updateStack
```

Display details about a specific database

```
# dbaascli database getDetails --dbname <db_name>
```

Get the list of PDBs in the database

```
# dbaascli database getPDBs --dbname <db_name>
```

Checks status of a database

```
# dbaascli database status --dbname <db_name>
```

Display Oracle Homes

```
# dbaascli system getDBHomes
```

Display Oracle Homes Details

```
# dbaascli dbHome getDetails --oracleHomeName <db_home>
```

List the details of all the Grid homes

```
# dbaascli system getGridHomes
```

List available DB/GI images

```
dbaascli cswlib showImages --product [GRID|DATABASE]
```

Create a database

```
# dbaascli database create -dbName <db_name> 1...
```

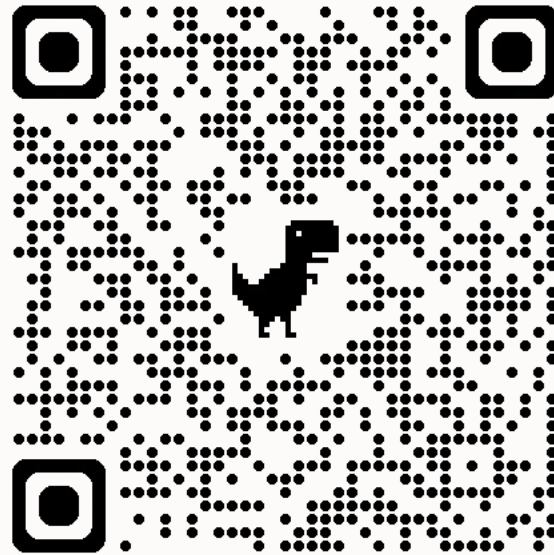
Database out-of-place patching

```
# dbaascli database move -dbname <db_name> 1...
```

¹... Not all parameters listed. Please check the command definition



Stay tuned



Operations Advisory
GitHub Repository



Thank you

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